

MIE 375 Fall 2025

PCB Mounting Assembly Manufacturing Project

Final Report

Enterprise Introduction (PM)

Over the course of the semester, Enterprise 05 successfully designed, manufactured, inspected, and assembled a printed circuit board mounting assembly that met all functional requirements set by the customer. The final assembly integrated perfectly with the provided baseplate, achieved a successful fit and alignment, and passed both the tip test and the shake test. Through the application of DFMA principles, the enterprise produced a manufacturable design capable of scaling to a projected production rate of 5,000 units annually. This report documents the complete development of the PCB mounting assembly, including design refinement, fabrication across multiple manufacturing methods, dimensional inspection, and final assembly. The enterprise worked with existing customer hardware, including but not limited to the base plate, dowel pins, and hex nuts. The fabricated components include the Clearance Mount, Standoff Mount, Sheet Metal Clips, Display Mounting Plate, and Sub-Assembly Fixture. Enterprise 05, led by project managers Luke Morrison and Benjamin Schaffer, consisted of 39 members, organized into 8 business units, each focusing on a distinct aspect of the project. Together, these units collaborated to manufacture all mounting components, assembly fixtures, inspection plans, and the final system.

1. Business Unit 01 Section

1.1. BU 01 Introduction

1.1.1. Introduction

BU01 was responsible for the design and production of the Clearance Mount using additive manufacturing. The Clearance Mount mates with the existing base plate and serves as a locating/mounting point for the Standoff Mount. The Clearance Mounts are manufactured using the MIS Innovation Shop Stratasys FDM printers.

1.1.2. Team and contribution

Name	Responsibilities	% Contribution
Cameron Siu	BU Lead, communication with the PMs and other BUs, Initial design	25
Marin Vigneras	CAD design, machine shop representative, Prototyping.	25
Samuel Judson	CAD design, drawing creation, printing and evaluating prototypes	25
Tenzin Sharchung	In process inspection, prototyping, design input, final part printing	25

1.2. BU 01 Methods and Procedures

1.2.1. Interpretation of specifications

The Clearance Mounts must be attached to the base plate without touching the interference plate. They must constrain the Standoff Mounts in the proper X, Y, and Z locations in relation to the

base plate. Finally, non-permanent cantilever snap joints must be included, as well as one hex nut and one heat-set insert.

1.2.2. Analysis & design decisions

This part underwent several iterations, primarily concerning the snap joints and the locating feature. This BU used online resources to assist in designing the snap joint geometry, and they discussed iterations with employees of the makerspace and machine shop. Working with BU02 and BU03, the locating feature was designed as a cross-shaped extrusion. This feature mates with square feet on the bottom of the Standoff Mount. One early design included a pin extrusion in the center of the part. This was determined to be infeasible due to its low expected durability. Lastly, the 82° Socket Flat Head Screw was chosen so that installation would leave it flush with the bottom of the base plate.

1.2.3. Fabrication and inspection

This design was printed on the Stratasys F123 Printers at the UMass Innovation Shop. ABS was used due to its flexibility and strength. The parts were printed on their sides to achieve high bending strength in the clips. The print lines ran parallel to the length of the clip, which minimizes shear failure between layers and allows elastic deformation. When the parts were printed, the support material was removed using a hot bath. This design did not need additional post-processing, and the heat insert was cleanly installed into the part. For inspection, the BU measured relevant dimensions with calipers and tested the installation to the Base Plate. Measurements of critical dimensions were taken using digital calipers, and this data was compiled into a database as seen in **Table 1**.

1.3. BU 01 Inspection results

1.3.1. Summary of results and their conformance to specifications

The in-process inspection indicated that the parts were within the required tolerance. However, the thorough inspection by BU08 revealed the parts to be slightly under tolerance in all dimensions. These discrepancies are due to the variability of 3D printing and shrinkage from cooling. Though the dimensions were not met, the process proved to be very controlled, as seen in the control charts, **Figures 2-4**.

1.4. BU 01 Discussion

1.4.1. Interpretation of inspection results.

All key dimensions of the part were out of tolerance. However, the functionality was unaffected. The only way to meet this tolerance requirement would have been to use a different process, which was not feasible for this project, given the 3D printing requirement. Improvements could be made by tweaking the dimensions in CAD to be slightly larger.

1.4.2. Performance vs prediction

These parts exceeded expectations. The biggest success was the cantilever clip design, which elastically deformed as expected and did not exhibit any plastic deformation after extensive testing. The Stratasys F123 printer can achieve a 0.008” tolerance, which means the process cannot meet the strict dimensional requirements that were imposed. However, the parts function properly, which suggests that tolerances are too strict. The cross-shaped extrusion also exceeded expectations, interfacing seamlessly with the Standoff Mount.

1.5. BU 01 Recommendations

1.5.1 For commercial manufacture and use of the assembly.

The best printer that the enterprise has access to, Stratasys F123, is not capable of consistently meeting the required tolerances. If production were scaled, this team would either need to adopt a different process, such as injection molding, or the customer would need to relax the tolerances. Relaxing tolerances is preferred as the parts function properly and cause no issues downstream.

1.5.2. For the class project

The beginning of the project was more chaotic and unorganized than it needed to be. Spending the first couple of weeks of class on the term project specifications would have cleared up much of the confusion resulting from the early deadlines and deliverables. Since BU01 was the first group to design and manufacture their parts, they had no reference frame for what was expected of them. Lastly, the Canvas page could have been more organized.

2. Business Unit 02 Section

2.1. BU 02 Introduction

2.1.1. Introduction

Business Unit 2 was responsible for the manual machining of the enterprise's Standoff Mounts. The goal of this step was to prepare the aluminum bar stock for BU03 to create complex features of the standoff mounts. This process considered DFMA principles to make machining efficient and repeatable.

2.1.2. Team and contribution

Name	Responsibilities	% Contribution
Abby Menendez	BU Lead, Upstream and downstream communication, organizing CAD files	30%
Manny Kum	Meeting minutes and QMS	17.5%
Jonathan Roussel	Deburring, organizing parts, and taking measurements	17.5%
Nate Hawkins	Milling lead and part drawings	17.5%
Owen Campbell	Deburring, organizing parts, and taking measurements	17.5%

2.2. BU 02 Methods and Procedures

2.2.1. Interpretation of specifications

The Standoff Mount is intended to raise the PCB and Display Mounting Plate to the correct Z location in the final assembly. In addition, the interface with the Clearance Mount had to be stable and reliable. Since the part was made in conjunction with BU03, the two groups had to work together to decide how the milling would be divided.

2.2.2. Analysis & design decisions

The design was driven by the milling equipment available to the enterprise. With DFMA in mind, the design does not include any external fillets or internal sharp corners. The design was created in tandem with BU01 and BU03 to ensure that everything was manufacturable and that the interfaces would be successful. Since this part is fabricated collaboratively with BU03, the features to machine were divided based on process capability.

2.2.3. Fabrication and inspection

This group manually machined five Standoff Mounts before final CNC machining by BU03. First, the aluminum stock was cut into five pieces on the band saw. Then face milling was conducted on the top and bottom of each part. Next, the parts were deburred, and the feet passes were made with the quarter-inch end mill. Using calipers to check relevant dimensions, all parts passed the first in-process preliminary inspection.

2.3. BU 02 Inspection results

2.3.1. Summary of results and their conformance to specifications

The inspection report of the Standoff Mount (STM) showed that all five parts failed. The only lengths manufactured by BU2 that passed inspection were 2.2 (z-axis) on STMs 2 and 3, as well as length 3.3 (z-axis) on STM 1. The rest of the lengths manufactured by BU2 were out of specification. The amount that each length was OOS was no more than a few thousandths of an inch, with the exceptions of 3.3 (z-axis) on STM5, which was OOS by 0.1526 inch (over) and 2.3 (y-axis), which was OOS by 0.2273 inch (over). These two dimensions, as well as all other OOS dimensions on measurement 3.3 (z-axis), were the only measurements that were OOS by being over length, rather than under.

2.4. BU 02 Discussion

2.4.1. Interpretation of inspection results.

Fabrication did not completely go as planned. When parts were measured in the machine shop, they matched the drawing, but the inspection report demonstrated that measurements were not precise enough. Many key dimensions were OOS. In addition, more machining was conducted than anticipated, as the legs on the Standoff Mounts did not initially mate with the Clearance Mount properly. These inaccuracies were likely a combination of human error and the process capability of the milling machines. Having parts improperly aligned with the Clearance Mounts was likely due to a tolerancing communication error between BUs 1 and 2.

2.4.2. Performance vs prediction

The final parts performed as expected. Though there were minor dimensional inaccuracies, the final assembly was not significantly jeopardized, and everything functioned properly. The prediction was that this process would be quite capable, but it proved to be less capable than anticipated. That said, it was very consistent.

2.5. BU 02 Recommendations

2.5.1. For commercial manufacture and use of the assembly.

To proceed with commercial manufacturing, the main priority would be limiting the total machining time of the part. It would be essential to create an assembly line with specialized machines for each step in the production process, as the setup and repositioning account for most of the time. It also takes time to switch bits out after each step of the process.

2.5.2. For the class project

The canvas could have been better organized. It was difficult to find specific information pertaining to the various business units. At first, all BUs had access to each other's assignments, which created undue stress and confusion. However, once each group could only see their own deliverables, this led to a feeling of disconnection from the rest of the team. Enterprise-wide meetings would have been beneficial, especially towards the beginning of the semester.

3. Business Unit 03 Section

3.1. BU 01 Introduction

3.1.1. Introduction

BU03 received the Standoff Mounts from BU02 and performed CNC machining to finish the parts for the final assembly. They worked with BU01, BU02, BU04, and BU05 to design the Standoff Mounts, as these parts were the cornerstone of the assembly. The Standoff Mounts were designed for manufacturability and assembly, and machining time was minimized.

3.1.2. Team and contribution

Name	Responsibilities	% Contribution
Brooke Oberlee	Coordinating with PMs, coordinating with other BUs, Delegating Tasks to BU Members, Design Alterations, Machining, Writing Meeting Minutes, Final Report	35%
Emmett Hughes	Machining, Writing Meeting Minutes, Final Report	15%
Caleb Ponder	Machining, Writing Meeting Minutes, Final Report	10%
Domenic Conidi	Design Alterations, Machining, Writing Meeting Minutes, Final Report	20%
Edward Zhou	Design Alterations, CAM, Machining, Writing Meeting Minutes, Final Report	20%

3.2. BU 03 Methods and Procedures

3.2.1. Interpretation of specifications

The feet and cantilever clip channels on the Standoff Mount interface with the Clearance Mount to provide the final placement for the PCB board in the X, Y, and Z planes. The Standoff Mounts are all 2.375" +/- 0.005" tall, so the final placement of the PCB board will be 3.625" above the base plate. The dowel pin hole ensures the PCB is aligned with and constrained by the four Standoff Mounts in the XY plane.

3.2.2. Analysis & design decisions

The hole diameter for the dowel pins was slightly undersized at 0.249” at a depth of 0.6” to achieve a press fit that would allow for the Sheet Metal Clips to clip onto the Standoff Mount. The cantilever clip channels were designed as 0.55” wide channels to allow for the cantilever clips to snap on without any interference. The top boss was extruded 0.1” to prevent the part from touching the bottom of the PCB. Curved pockets on the top of the part were added to allow for automatic assembly while maintaining the 2” by 1.875” base.

3.2.3. Fabrication and inspection

The top boss was milled with one pass of 0.1”, and the side pockets were milled in 8 passes, 7 being at 0.1” and the final pass being 0.05”. The robot arm indentations were milled with 5 passes, each at 0.1”. The cantilever clip channels were milled in 2 passes, each at the same depth of 0.075” to get the width of 0.55”. The dowel pin hole was drilled to a diameter of 0.246” and then reamed to 0.249”. After each part was manufactured, key features were measured using calipers.

3.3. BU 03 Inspection results

3.3.1. Summary of results and their conformance to specifications

For all parts, the CNC path was set to be as close to nominal as possible. After machining, calipers were used to measure important dimensions of each part. At first glance, the caliper measurements indicated that the parts were in tolerance, but this was disproven after final inspection. The inspection results were returned with every part as a failure to pass. There is at least one notable error, with STM 5’s measured cutout at 0.9 instead of 0.75 in. This was due to an improperly tightened mill bit that slipped after each pass of the machine, resulting in the

deeper cutout. Many tolerance errors could be fixed by locking the piece better, ensuring the bit is positioned properly, and/or removing chips and pieces. However, most tolerance issues seem to stem from measurement and improper tolerances.

3.4. BU 03 Discussion

3.4.1. Interpretation of inspection results.

As reported above, the official inspection status was a failure. This was likely due to operator error, as the process is capable. Certain measurements, such as circularity, could be increased to $\pm 0.01''$ or ± 0.10 in. Some tolerances were mislabeled as $\pm$ instead of $+/-$. Lastly, a small number of measurements failed due to operator error, either on the machining side, with the occasional visibly noticeable mistake, or on the inspection side, such as measurements being returned as more than a full inch off when they are visibly not.

3.4.2. Performance vs prediction

Through preliminary assembly, the part's overall performance met design intent. After adjacent business units resolved initial interference and manufacturing issues associated with less precise processes, the part functioned as intended. The snap joints demonstrated sufficient strength, and no assembly issues were observed.

3.5. BU 03 Recommendations

3.5.1. For commercial manufacture and use of the assembly.

There could be a chamfer at the bottom of the part to assist the UR5e robot in guiding the cantilever clip joints into the channels milled on the sides. When designing the part, the width of the cantilever cutout matched the exact size of the clip joint itself, which caused issues with the assembly. For future manufacturing, the channel should be designed and milled significantly larger than the clip, so there are no issues fitting the two together. Lastly, the hole for the dowel pin needs to be slightly deeper in future iterations, so that the Sheet Metal Clips can be removed with less effort.

3.5.2. For the class project

For this term assembly project, having leads for design, calculations, machining, etc. would be helpful in making sure that each member of the group does their share of the work and the business unit lead doesn't have to pick up the slack of other group members. The division of labor in this group was sub-optimal, because it is difficult to ensure everyone does an even share.

4. Business Unit 04 Section

4.1. BU 04 Introduction

4.1.1. Introduction

BU04 was responsible for designing and manufacturing the Sheet Metal Clips and Display Mounting plate. The clips fix the display plate and PCB board to the Standoff Mount, while the display plate connects to the fan and display. This group used the pan brake and water jet in fabrication.

4.1.2. Team and contribution

Name	Responsibilities	% Contribution
Sarah Bryan	BU Lead, Communication with PMs and other BU leads, task delegation, presentations, fabrication, CAD	35%
Nicolas Brooks	Project Research, Presentation, Fabrication	20%
Danial Mirza	Project Research, Presentation, Fabrication	15%
Tung Lam	Project Research, Presentation, Fabrication, CAD modifications	15%
Andrew Klein	Project Research, Presentation, Fabrication	15%

4.2. BU 04 Methods and Procedures

4.2.1. Interpretation of specifications

BU4 is responsible for fabricating both mounting clips and the Display Mounting Plate. The base plate will be located by the dowel pins in the X and Y direction and cannot touch the PCB in any of the no touch zones. Both the clips and plate must be manufactured using the water jet and pan brake. The plate fan and display could not touch the PCB anywhere in the no-touch zones.

Lastly, there also had to be at least 50% of the fan area cut out with a maximum opening of 0.3125”.

4.2.2. Analysis & design decisions

Multiple vent prototypes were proposed, and an ideal design and location were chosen. Since the mounting plate had to be located on the base plate using dowel pins, it was decided to use pin holes on one side of the plate and slots on the other. Manufacturing engineering analysis significantly impacted the designs because most of the part manufacturing was done manually. Furthermore, the team was very cautious about the spring back of the material during the process of sheet metal bending. It was important to make sure all the bends were manufacturable within the pan brake and not collide in the order of operations.

4.2.3. Fabrication and inspection

BU04 used the pan brake to bend the clips. Prior to bending, a deburring process was carried out to remove any sharp edges, and a caliper was used to carefully mark the bend locations on the parts. After the sheet metal bending processes, the parts were passed for assembly. The in-process inspection focused on measuring the lengths between bends using calipers, and the bend angles using a protractor.

4.3. BU 04 Inspection results

4.3.1. Summary of results and their conformance to specifications

The material succumbed to spring back which caused bend angle variation. The parts were typically overbent in a 2-5° manner. None of the final clips showed twists, unintended curvatures, or plastic deformation. The pin holes were enlarged to resolve any interferences with BU2’ and BU3’ parts which measured consistently larger than the original prototype.

Measurement of the clip lengths and bend-to-bend distances occasionally exceeded the ± 0.01 in tolerance. However, this error was expected due to the by hand processing. Both parts met functional requirements, so the fact that they didn't pass inspection shows that the inspection requirements are too strict.

4.4. BU 04 Discussion

4.4.1. Interpretation of inspection results.

The parts were completely functional in the overall assembly. Although the final dimensions of the components exceeded many of the tolerances, no issues were found during the assembly. Therefore, dimensional inaccuracies are not enough to cause any serious issues. The pan brake has inherent variability to it because of operator error, so besides choosing a different process there is not much that can be done.

4.4.2. Performance vs prediction

The sheet metal parts performed as predicted. Since the bending processes were measured and completed by hand, human error was a significant factor in the inaccuracies found in the final product. Fortunately, a lot of the challenges that were faced as a team were mostly handled in the initial stages of the bending process, which led to a smooth process for actual manufacturing.

4.5. BU 04 Recommendations

4.5.1. For commercial manufacture and use of the assembly.

The hand brake was prone to human error. A more efficient manufacturing process is sheet metal pressing. For bulk production, this method not only reduces the time required per component but also significantly lowers labor costs.

4.5.2. For the class project

From the perspective of the students, the MIE375 project has been both a challenge and a learning experience. The communication between the enterprise was challenging at times given the number of people working together. It would be beneficial to include a formal meeting with the entire enterprise during class time so they can better communicate with one another.

5. Business Unit 05 Section

5.1. BU 05 Introduction

5.1.1. Introduction

BU05 focused on designing and manufacturing the assembly fixture that is used to orient and locate the Clearance Mount and Standoff Mount for assembly by the Universal Robot UR5E. The final design consists of a two-half interlocking system that attaches to the assembly peg board with split dowel pins. This part was fabricated using the Stratasys 3D printers.

5.1.2. Team and contribution

Name	Responsibilities	% Contribution
Arjun Gandham	Team Lead, CAD Designer, Prototype Iterations	35%
Eric Lo	CAD Designer, Prototype Iterations	27.5%
Naoya Miyamoto	CAD Designer, Prototype Iterations	27.5%
Theo Turlentes	Report Writing	5%
Rider Bumgarner	Report Writing	5%

5.2. BU 05 Methods and Procedures

5.2.1. Interpretation of specifications

The assembly fixture must hold and orient the Clearance and Standoff Mounts by restricting movement in the X, Y, Z, and rotational axes. It must be manually installed onto a ¼” x 18” x 24” HDPE pegboard with 3/16” holes on 1” hole spacing and must use split dowel pins. The design requires two halves, attachable by joint or snap-fit connection. The fixture must be fabricated using Stratasys FDM printers. During design, the team must optimize the weight and make sure that the assembly process is as smooth as possible.

5.2.2. Analysis & design decisions

This team was mainly concerned with functionality, with weight being a secondary consideration. The Stratasys F123 printers were used for reliability. The team decided on a T-slot connection because it is sturdy and reliable. Early concepts included tall, thin walls to hold each of the mounts, but through the iterations they were removed to save weight.

5.2.3. Fabrication and inspection

Prototyping began in the makerspace with test prints to evaluate the design. Once the team was satisfied, the final design was printed on the Stratasys F123 printers. Some post-processing was necessary to remove residual material. The group realized that the dowel pins were not in an optimal spot for assembly, so holes were drilled in the proper spot and dowels were installed. After taking dimensions by hand, the parts appeared to match the drawings well enough to fit the parts as seen in Table 2.

5.3. BU 05 Inspection results

5.3.1. Summary of results and their conformance to specifications

After production, the fixture was tested extensively to see if it could perform as expected. After multiple iterations with the T-slot, mount interfaces through test prints, and spring pin connection dimensions, everything functioned as expected. There were no issues with any of the connections, and the fixture was passed off to BU06 for robotic assembly.

5.4. BU 05 Discussion

5.4.1. Interpretation of inspection results.

The fixture was not inspected, because BU08 and the PMs decided it would have been unnecessary. The fixture functioned perfectly; it constrained the Clearance Mount and Standoff

Mount to allow the Co-bot ample space to grab and attach them as needed. Since there were no fixture tolerance requirements from the customer, inspection was skipped to save time.

5.4.2. Performance vs prediction

The fixture performs exactly as predicted. Every piece fits together properly and constrains the Standoff Mount and Clearance mount as needed. BU06 uses the fixture to perform the robotic assembly with no issues, and more of these could be made to scale production.

5.5. BU 05 Recommendations

5.5.1. For commercial manufacture and use of the assembly.

For commercial manufacturing, there are several aspects that could be improved upon. We would have the fixture entirely 3D printed instead of having dowel pins for the Clearance Mount. We also could have significantly reduced the fixture weight by hollowing out the middle section. The critical components of this fixture are the pockets for the mount and holes and the bottom.

5.5.2. For the class project

This project could have been more challenging. More functional requirements or the requirement of an assembly fixture would have allowed a larger impact on the project. The team was only working with BU02, BU03, and BU06. Otherwise, they were not able to collaborate with other units.

6. Business Unit 06 Section

6.1. BU 06 Introduction

6.1.1. Introduction

BU06 was responsible for programming the UR5E Co-bot arm to automate the assembly of the Clearance Mount and Standoff Mount using the fixture from BU05. The team was successful and met the functional requirements.

6.1.2. Team and contribution

Name	Responsibilities	% Contribution
Evan Sousa	BU Lead, communication lead	25%
Rich Wirtanen	Coding lead, testing	25%
Colin O'Brien	Implementation of plans, equipment changeover times and process flowcharts	25%
Michael Farber	Presentation	25%

6.2. BU 06 Methods and Procedures

6.2.1. Interpretation of specifications

The assembly must be completed using the UR5e robot. The robot must locate, orient, and assemble the Standoff Mount to the Clearance Mount. The robot constraints are an insertion force limit of 25N, a Base, Joint, and Elbow speed limit of 100 degrees per second, and a speed limit of 150 degrees per second for Wrists 1, 2, and 3. The Components were to be placed in staging positions on the fixture before assembly.

6.2.2. Analysis & design decisions

For assembly, this team decided which method was most time-efficient and repeatable, ultimately opting to manually program the robot rather than using the Palletizing Wizard or the Wrist Camera. The team chose this because it was the fastest to implement. It was also

determined that the Wrist Camera would be less efficient due to the time spent “finding” each component and determining part orientation.

6.2.3. Fabrication and inspection

First, the fixture was aligned on the pegboard. The Standoff Mounts and Clearance Mounts were then placed onto the fixture. The assembly program is then run on UR5e. The robot is moved to the first waypoint with open grippers and closes the grippers to grab the Standoff Mount. Next, the robot moves to a waypoint above the Clearance Mount before moving straight down. When the robot moves down, the Standoff Mount interfaces with the Clearance mount. The grippers are then opened, and the robot returns to the first waypoint. Once all parts are assembled, they are unloaded and quickly inspected to ensure all parts are assembled properly.

6.3. BU 06 Inspection results

6.3.1. Summary of results and their conformance to specifications

The time to power on and set up the robot was approximately 2 minutes and 38 seconds. The assembly time itself took 28 seconds and had few errors in its movements. The robot takes the shortest path, minimizing process time. The position of the robot arm is far from the mounts at the first waypoint to give operators plenty of space. The in-process visual inspection phase of the assembly took 31 seconds. This process checked for connectivity errors between parts.

6.3.2. Analysis of Outliers

The outliers occurred because of the gap between the base of the clearance mounts and the fixture. The clearance was too large, which allowed the mounts to wiggle slightly in their slots. As a result, they could rotate just enough to occasionally get caught underneath the Standoff Mount while the robot was trying to connect them. This issue is considered an outlier because it

happens very rarely. When the parts are placed correctly, flat against the fixture wall on the side closest to the operator, it does not occur. The problem only arises if the robot stand is bumped or if the pieces are positioned incorrectly.

6.4. BU 06 Discussion

6.4.1. Interpretation of inspection results.

According to the inspections completed by BU08, the key dimensions of sub-components were slightly out of specification relative to the design intent. All five Clearance Mounts and all five Standoff Mounts exhibited multiple out-of-tolerance measurements, and the Standoff Display Mount exceeded tolerance limits in every measured parameter. In total, each sub-component failed at least four and up to ten critical dimensions. These findings aligned with the team's early observations during programming, where dimensional mismatches prevented smooth insertion of the Standoff Mounts into the Clearance Mounts. Based on these results, the team identified the dimensional issues that required correction before programming could proceed.

6.4.2. Performance vs prediction

The robot functioned exactly as predicted, and the code ran smoothly. Initially, the Clearance Mount and Standoff Mount did not lock properly, thereby delaying assembly. The respective business units resolved the issue promptly. Despite this delay, the team adhered to deliverable deadlines to keep the whole enterprise on track.

6.5. BU 06 Recommendations

6.5.1. For commercial manufacture and use of the assembly.

The tolerances between the fixture and the mounts could be tweaked slightly by making changes to the CAD, in addition to the tolerances between the mounts themselves. This would eliminate the few outliers observed in testing. However, this could be unnecessary given how infrequently this issue arose. Aside from that, the process was efficient and could feasibly be scaled up to meet demand.

6.5.2. For the class project

This team believes that Canvas could have been more organized. Another problem was that at the beginning of the semester, assignments and due dates for each business unit were visible to all students, which caused confusion and unnecessary stress. Maybe an outside resource or site can be used to track individual business unit progress for smooth progression.

7. Business Unit 07 Section

7.1. BU 07 Introduction

7.1.1. Introduction

The goal of BU07 was to streamline the assembly process so production can be scaled up. The team communicated with upstream BUs to make sure all parts were designed with assembly in mind. Once parts were fabricated, BU07 created efficient assembly instructions.

7.1.2. Team and contribution

Name	Responsibilities	% Contribution
Matthew Bressler	Building Assembly, making presentation, Final Report, and flow chart	25%
Wassim Omalek	BU Lead, communication leader, Final Report, work instructions	25%
Matthew McCoy	Work instructions, final report, and flow chart	25%
William O'Shea	Double checking, making presentation, final report, and flow chart analysis	25%

7.2. BU 07 Methods and Procedures

7.2.1. Interpretation of specifications

For the final assembly, the fan must be installed at the underside of the Display Mounting Plate, the display panel must also be installed, nothing must touch the PCB “NO-TOUCH” zones, and the team cannot remove or loosen the Interference Plate during assembly. The team is allowed access to the bottom of the Base Plate during assembly.

7.2.2. Analysis & design decisions

The team had to decide on assembly methods that minimize time for assembly in accordance with restrictions and procedure requirements. This included flipping the assembly to screw into

the heat inserts first, before flipping the assembly back over and screwing into the hex nuts. This was done because the lack of visibility of the bottom of the plate when screwing into the hex nuts caused human error of accidentally screwing into the heat insert slot. By screwing the heat insert first, there is no risk of accidentally screwing into them while meaning to screw into the hex nut. The team also had to decide on the most efficient method of screwing the fan in, which involved using both the Allen wrench to screw the fastener and pass-through-wrench to hold the nylon insert hex nut while screwing. This was all done while holding the display plate on its side to ensure the screws did not fall out while fastening. Screwing the fan into the display plate while it was right side up was not possible due to the lack of space under the plate, making the side-ways orientation necessary.

7.2.3. Fabrication and inspection

The team installed the dowel pins into the top of the Standoff Mount, then installed the sub-assemblies to the base plate. Then, the assemblers bolted in the fan and secured the Go-No-Go Gage on the Display Mounting Plate. The team installed the PCB board onto the subassemblies, then installed the Display Mounting Plate on top. The assemblers then secured the Display Mounting Plate to the Standoff Mount using a sheet metal clip in each corner.

7.3. BU 07 Inspection results

7.3.1. Summary of results and their conformance to specifications

12 of the dimensions were out of specifications, 7.1-X is -0.0045 in out of tolerance, 7.2-Y is 0.0295 in out of tolerance, 7.3-Z is -0.0151 in out of tolerance, 7.4-Z is -0.0265 in out of tolerance, 7.5-Z is -0.0461 in out of tolerance, and 7.6-Z is -0.2469 in out of tolerance. For 7.1

right side, flatness is 0.0014 in out of tolerance; parallelism A is 0.2769 in out of tolerance. For 7.5 on the right side, flatness is 0.0324 in out of tolerance; parallelism C is 0.0315 in out of tolerance. For 7.6 left side, flatness is 0.0275 in out of tolerance; parallelism C is 0.0094 in out of tolerance. It can be seen here that all 12 of these dimensions do not conform to specifications. 3 dimensions are within specifications, 7.1 right side perpendicularity A passed inspection with a dimension of 0.0043 in. For 7.4 left side, flatness passed inspection with a dimension of 0.0014 in; parallelism C passed with a dimension of 0.0234 in. Only 3/12 of the actual dimensions were within tolerance, which leads the inspection status to be “fail”.

7.4. BU 07 Discussion

7.4.1. Interpretation of inspection results.

The failing of inspection on the final assembly stems from the variation found in the various manufacturing methods of previous business units. The dimensions of the assembly are below and above tolerance for the various dimensions; however, the deviations from nominal dimensions are small as seen in the inspection result summary. Although the dimensions did not meet dimensional requirements, the functionality of the assembly was not compromised.

7.4.2. Performance vs prediction

The team predicted that the assembly would rattle or fall apart when undergoing the shake test. However, the assembly exceeded expectations; it passed the shake and tip tests. The high performance juxtaposed with the failed inspection shows that certain tolerances may be tighter than what is actually necessary for a functional assembly.

7.5. BU 07 Recommendations

7.5.1. For commercial manufacture and use of the assembly.

Including an assembly fixture to hold the sub-assemblies to the base plate would drastically decrease assembly time. During assembly, parts could sometimes wobble or move out of place. Also, the hardest part of assembly was attaching the fan to the display plate, so another fixture to hold the nuts in place would save at least one minute per assembly.

7.5.2. For the class project

Having required meetings, either between specific BUs or the entire enterprise, would have made the process much smoother. BU07 was not in the loop with regards to the CAD from earlier BUs as much as they should have been, which would have been less of a problem if enterprise-wide meetings were set up through class time. Also, there could have been more to do towards the beginning of the semester.

8. Business Unit 08 Section

8.1. BU 08 Introduction

8.1.1. Introduction

BU08 inspected the individual parts and the final assembly to check if the parts were within the desired tolerances and matched the engineering drawings. Using the FARO Arm Gage, the team evaluated critical dimensions to determine if Enterprise 05 met the dimensional requirements of the project.

8.1.2. Team and contribution

Name	Responsibilities	% Contribution
Danielle Stalmon	BU Lead, communication lead, meeting minutes, organizing files, FARO Arm operator, inspection report generator	85%
Devin Rego	Scheduling meetings, assisting with inspection	5.5%
Patrick McDonald	Scheduling meetings, assisting with inspection	5.5%
Tanner Dweck	Presentation, assisting with inspection	3%
Christopher Machado Martinez	N/A	1%

8.2. BU 08 Methods and Procedures

8.2.1. Interpretation of specifications

The team's role in this project was to inspect the parts and confirm that they would function as intended. They developed inspection plans tailored to the different part geometries, and after completing the inspections, they analyzed the results to identify which processes could be scaled up for mass production.

8.2.2. Analysis & design decisions

The inspection team used the engineering drawings to select datums, then decided on the optimal part orientation. They planned the measurement sequence to be efficient and accurate. Some parts had to be elevated to access all six planes. They decided not to inspect the fixtures because the fixtures passed a go-no-go test and functioned well.

8.2.3. Fabrication and inspection

First, this team reviewed the drawings to ensure that all required dimensions were included. After this confirmation, steps were taken to maintain strict organization. This included color-coordinating inspection hexagons to the sides of the part on which the measurement would be taken. After setting up the FARO Arm, parts were mounted in clay, and the inspection was performed. Finally, full inspection reports were created using the gathered data. Detailed work instructions can be found in **Figure 1**.

8.3. BU 08 Inspection results

8.3.1. Summary of results and their conformance to specifications

Clearance Mounts (CLM) 1–5 were inspected for lengths 1.1, 1.2, and 1.3, along with positional and perpendicularity GD&T, as shown in **Table 3**. Although all parts failed overall inspection, Plane 3's perpendicularity to datum A was the most consistent measurement. CLM 2 was closest to meeting specifications due to minimal variation. No major outliers were observed; deviations were primarily associated with length 1.3 and were small enough to indicate minor manufacturing variation. Standoff Mounts (STM) 1–5 from Business Units 2 and 3 were inspected for lengths 2.1–2.3 and 3.1–3.3, along with positional, flatness, perpendicularity, and parallelism GD&T, as summarized in **Table 4**. While all parts failed overall inspection, length

3.1 was the most consistent across all parts, and STM 1 most closely met specifications. Measurement 2.4 for STM 1 showed the largest deviation and is likely due to inspection error, as all other out-of-tolerance values were small and consistent with minor manufacturing variation. Sheet Metal Clips 1–6 were inspected for lengths 4.1–4.3, perpendicularity, and angularity GD&T, as shown in **Table 5**. Although all clips failed inspection, plane C’s perpendicularity to datum B and the angularity to datum B were the most conforming features. The clips exhibited greater variation than other components, likely due to hand-bending and pre-inspection testing. The Display Mounting Plate was inspected for lengths 4.4–4.6, perpendicularity, and parallelism GD&T, with results shown in **Table 6**. This part failed all measurements, with all features exceeding tolerance. As with the clips, hand-bending contributed to dimensional inconsistency that could impact final assembly fit.

The final assembly was inspected for lengths 7.1–7.6, perpendicularity, and parallelism GD&T, as shown in **Table 7**. Measurements were inconsistent, with many dimensions under tolerance, likely due to tolerance stack-up from individual components. Despite dimensional variation, overall fit was achieved. While all features were inspected, certain dimensions were more critical, such as length 1.2 on the Clearance Mount, which directly affected overall assembly height and alignment.

8.4. BU 08 Discussion

8.4.1. Interpretation of inspection results.

After each inspection was completed, the results and corresponding inspection steps were uploaded to the group folder so that all business units, particularly the unit responsible for

manufacturing the part, could review measurement discrepancies. Each inspection was conducted with a predefined plan; however, minor adjustments were occasionally required due to equipment constraints. Once final drawings were issued, the only allowable modifications were to the inspection approach, as no changes to part geometry were permitted.

8.4.2. Performance vs prediction

There was a sharp learning curve for the FARO Arm. The first inspection was a challenge as it was the first time the team members used the software without help. Inspections took much longer than anticipated. There were also software issues midway through the project, which created stress. Although there were quite a few out of tolerance dimensions for each part, they were small enough that they could be disregarded because they didn't impact the assembly.

8.5. BU 08 Recommendations

8.5.1. For commercial manufacture and use of the assembly.

To meet the tolerance requirements, the enterprise must either choose more capable processes, provide more training to the enterprise's machine operators, or slightly relax tolerances. By-hand machining and bending proved to have the greatest errors, which must be corrected. Certain non-critical dimensions that were out of tolerance must be re-evaluated to see if the group can be less strict with them.

8.5.2. For the class project

The distribution of work across BUs could have been much better. BU08 had significantly more work to do than most of the other BUs. Finding ways to hold people accountable during the semester would be beneficial.

9. Enterprise Conclusion (PM)

Over the course of the semester, Enterprise 05 was largely successful in meeting the project objectives. The final assembly, shown in **Figure 14**, satisfied all functional requirements, and strong communication between business units supported effective application of DFMA principles. On the day of assembly, the system passed both inversion and shake testing, demonstrating robust and foolproof functionality. While many team members entered the project with limited manufacturing experience, the challenges encountered contributed to meaningful learning and skill development.

The primary challenge faced by Enterprise 05 was meeting dimensional tolerances, particularly when machining metal components using manual equipment such as the mill and pan brake. Despite these limitations, the components performed as intended, and **Table 7** shows that the final assembly achieved sufficient dimensional accuracy to ensure proper fit and function. Overall, the results demonstrate that the design and manufacturing approach could reasonably scale to the target production rate of 5,000 units per year.

The Enterprise was largely in agreement about potential future modifications to the project. For instance, there was not enough guidance at the beginning of the semester, given how early the deadlines were. The workload distribution between BUs was often uneven, and within the BUs, there were many individuals who did not contribute, leading to division of labor concerns. Overall, the team was satisfied with the project and its real-world application and would not suggest major changes.

Appendix

Table 1 – In-Process Inspection Data Business Unit 01

		Part Number										
		1		2		3		4		5		
Cantilever Snap Joints	Clip location:	In	Out	In	Out	In	Out	In	Out	In	Out	Expected
	Width	0.501	0.503	0.502	0.501	0.501	0.502	0.502	0.500	0.502	0.503	0.500
	Thickness	0.081	0.082	0.079	0.080	0.083	0.083	0.082	0.081	0.077	0.080	0.075
	Head Thickness	0.141	0.138	0.141	0.140	0.143	0.141	0.142	0.141	0.140	0.141	0.145
	Overhang Length	0.060	0.056	0.062	0.060	0.061	0.058	0.060	0.060	0.063	0.062	0.065
Standoff Mount Location (greater measurement of two sides)	Interference Plate to Top (exclude Top Extrusions or clip)	0.344		0.3425		0.342		0.34		0.34		0.3438
	Base Plate to Top (exclude Top Extrusions or Clip)	1.25		1.25		1.248		1.247		1.245		1.25
	Short Width of Top Extrusion	0.586		0.585		0.5855		0.583		0.5835		0.585
	Long Width of Top Extrusion	1.228		1.224		1.226		1.231		1.232		1.23
Base Plate Extrusion Properties	Hex Nut Can be Screwed?	Yes		Yes		Yes		Yes		Yes		
	Fits in Base Plate	Yes		Yes		Yes		Yes		Yes		

Table 2 – In-Process Inspection Data Business Unit 05

Feature	Solidworks Length(inches)	By-Hand Dimensioning(inches)
Bottom Hole	0.1875	0.182
Clearance Mount Depth	0.13	0.1235
Clearance Mount Pin Diameter	0.25	0.2395
Clearance Mount Pin Height	0.5	0.417
T-Slot Clearance	0.01	0.095
Standoff Mount Height	0.125	0.1312

Table 3 – Clearance Mount Inspection Data (Inches)

	1.1 (Y-Axis)	1.2 (Y-Axis)	1.3 (X-Axis)	Plane 3 $\perp$ to datum A	Bigger circle ϕ to datums A, B, &C	Smaller Circle ϕ to datums A, B, &C
CLM 1	-0.0042	-0.0085	-0.0004	Pass	2.7324	1.1997
CLM 2	-0.0066	-0.0004	Pass	Pass	0.0009	0.0984
CLM 3	-0.0051	-0.0044	-0.0032	Pass	2.7241	1.1890
CLM 4	-0.0044	-0.0014	-0.0041	Pass	2.7118	1.1741
CLM 5	-0.0070	-0.0076	Pass	Pass	2.7350	1.2034

Table 4 – Standoff Mount Inspection Data (Inches)

	2.1 (X-Axis)	2.2 (Z-Axis)	2.3 (Y-Axis)	3.1 (X-Axis)	3.2 (Y-Axis)	3.3 (Z-Axis)	Datum D ϕ to Daums A,B,&C	$\square$ of side 2.1	Plane 2 $\perp$ to datum C	Top bor // to datum E	Bottom bor // to datum F
STM 1	-0.0068	-0.0057	0.2273	Pass	Pass	Pass	0.3654	Pass	0.0086	Pass	Pass
STM 2	-0.0059	Pass	-0.0052	Pass	Pass	0.0059	0.3468	0.0088	0.0008	0.0016	Pass
STM 3	-0.0033	Pass	-0.0125	Pass	Pass	0.0052	0.1013	0.0039	0.0003	Pass	0.0804
STM 4	-0.0092	-0.0022	-0.0022	Pass	Pass	0.0054	0.8214	0.0201	0.0043	Pass	Pass
STM 5	-0.0061	-0.0025	-0.0026	Pass	-0.2334	0.1526	0.0778	0.0177	0.0037	0.0025	0.0117

Table 5 – Sheet Metal Clips Inspection Data (Inches)

	4.1 (Y-Axis)	4.2 (X-Axis)	4.3 ($\angle$)	Plane $\perp$ to datum A	Plane C $\perp$ to datum B	Plane B $\perp$ to datum A	Plane $\angle$ to datum B
SM Clip 1	0.2252	0.0259	-0.55°	0.0243	Pass	0.0126	Pass
SM Clip 2	-0.0594	-0.0266	7.18°	0.0085	Pass	0.0058	Pass
SM Clip 3	0.1002	-0.0044	-0.24°	0.0203	Pass	0.0036	Pass
SM Clip 4	0.1439	0.0377	8.6°	0.0190	Pass	0.0055	Pass
SM Clip 5	0.1692	-0.0426	1.10°	0.0203	Pass	Pass	Pass
SM Clip 6	0.1792	-0.0260	-1.48°	0.0037	Pass	0.0252	Pass

Table 6 – Sheet Metal Display Plate Inspection Results (Inches)

	4.4 (Z-Axis)	4.5 (X-Axis)	4.6 (X-Axis)	Plane B $\perp$ to datum C	Plane $\perp$ to datum B	Plane $\perp$ to datum C	Plane // to datum A
SM Display Mount	0.0897	0.2664	0.0185	0.0332	1.2869	0.0349	0.0534

Table 7 – Final Assembly Inspection Results (Inches)

	7.1 (X-Axis)	7.2 (Y-Axis)	7.3 (Z-Axis)	7.4 (Z-Axis)	7.5 (Z-Axis)	7.6 (Z-Axis)	7.1 R Side ⊥ to datum C	7.1 R Side // to datum C	7.6 L Side // to datum C	7.4 L Side // to datum C
Final Assembly	0.0045	0.5569	0.0151	0.0265	0.0461	0.2469	Pass	0.2769	0.0094	Pass

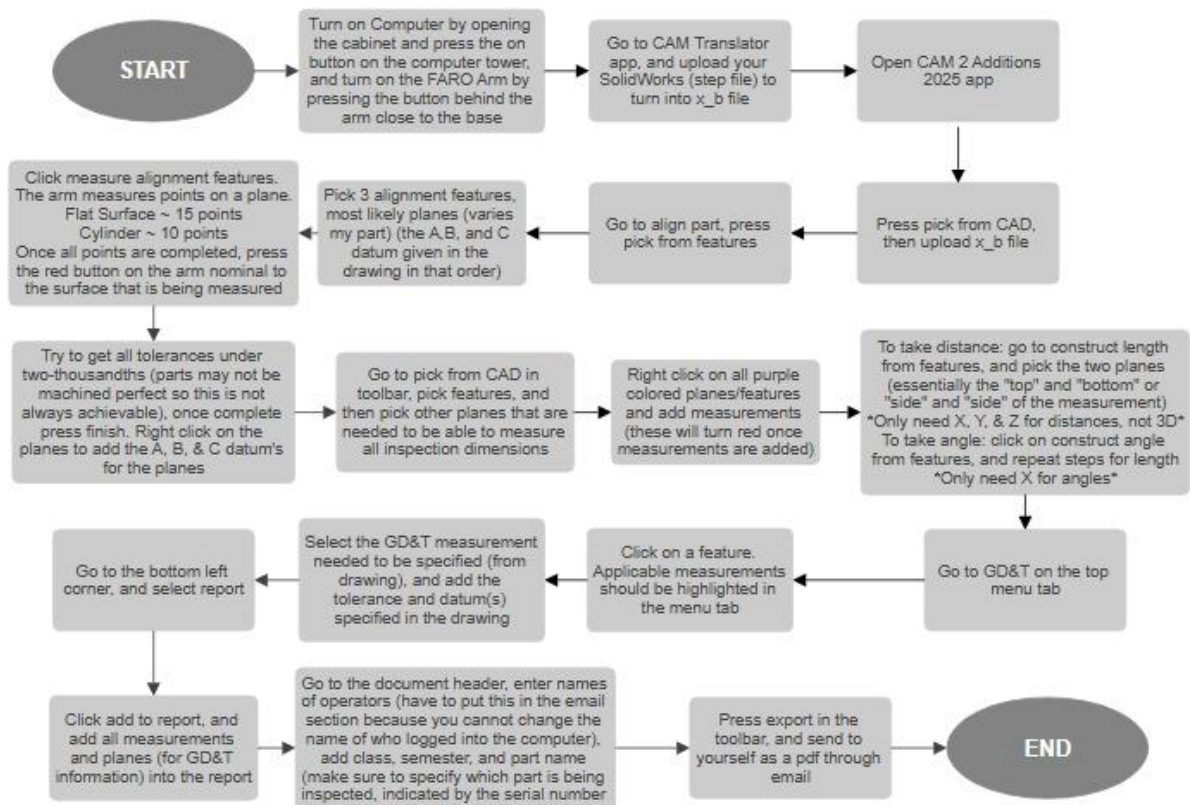


Figure 1 – Business Unit 08 Inspection Work Instructions

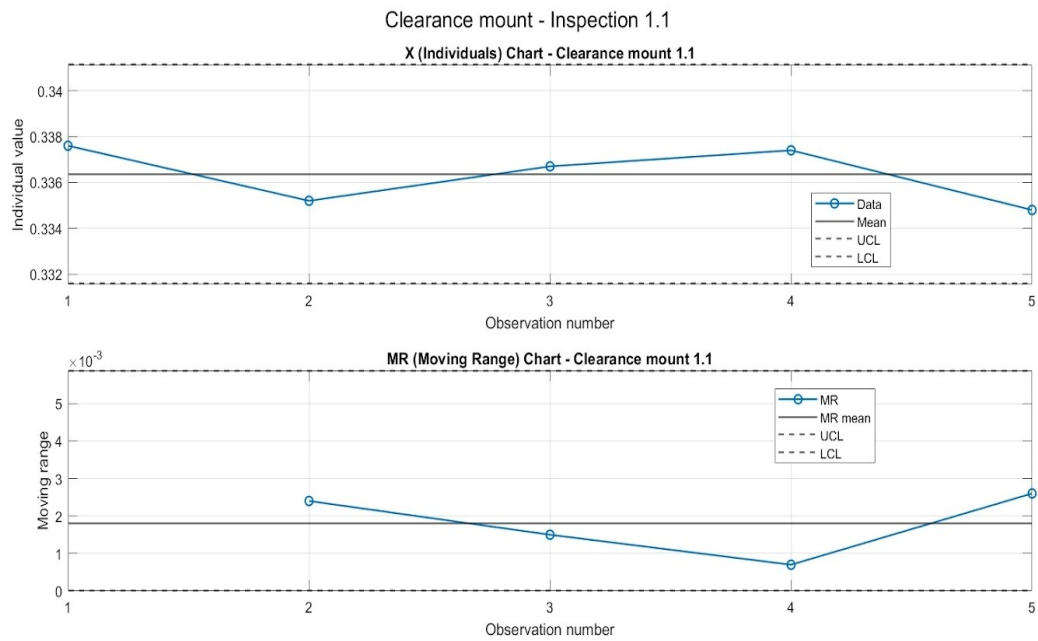


Figure 2 – Clearance Mount 1.1 Control Chart - XMR

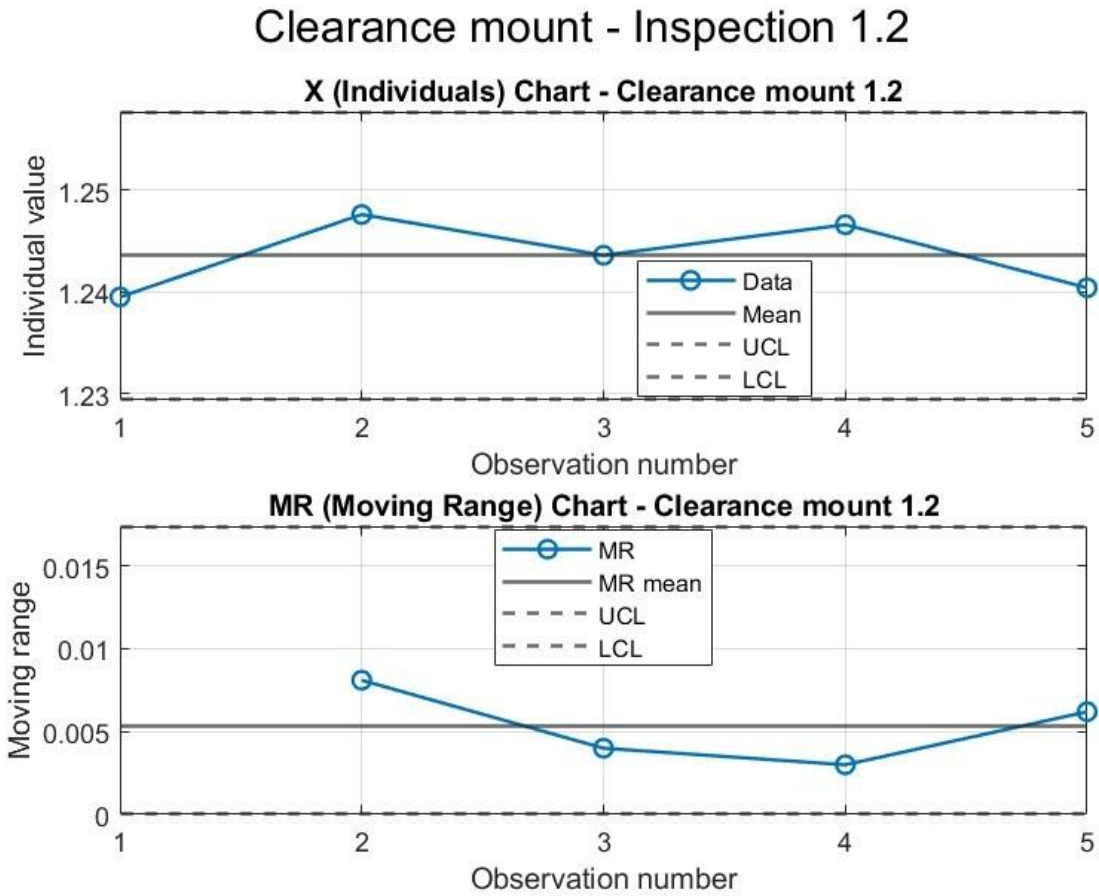


Figure 3 – Clearance Mount 1.2 Control Chart - XMR

Clearance mount - Inspection 1.3

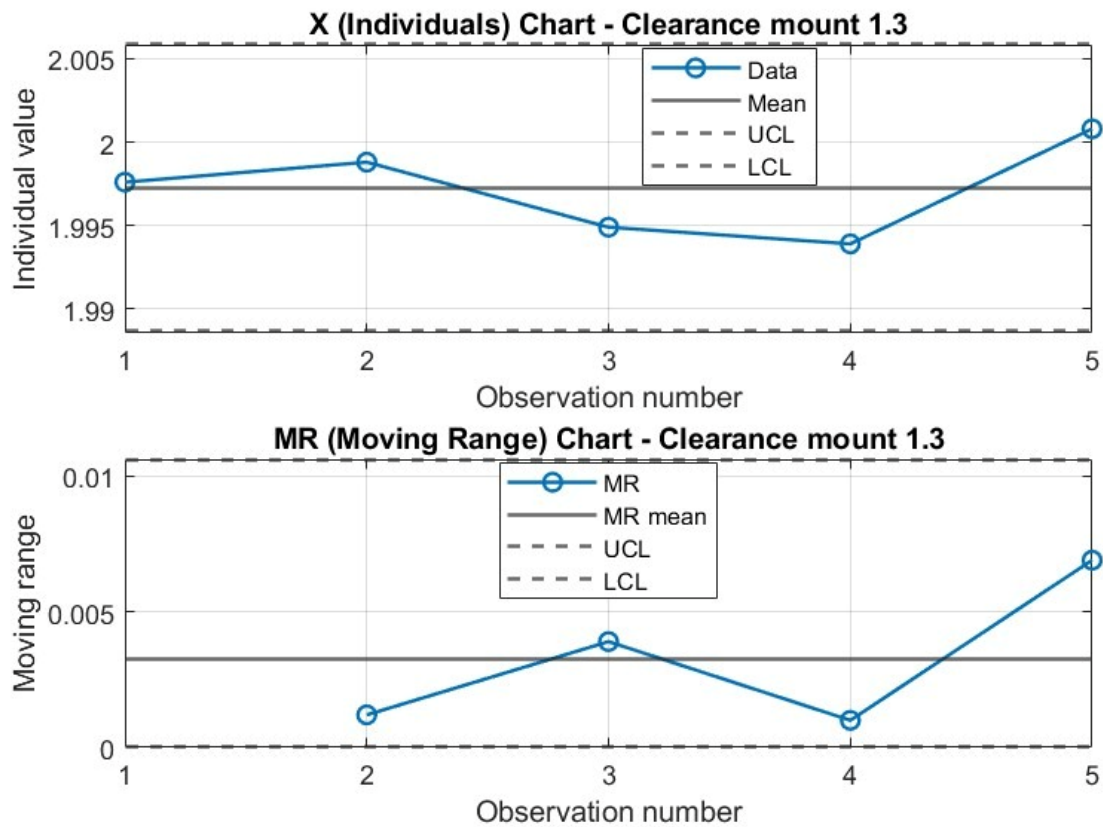


Figure 4 – Clearance Mount 1.3 Control Chart - XMR

Standoff mount (manual) - Inspection 2.1

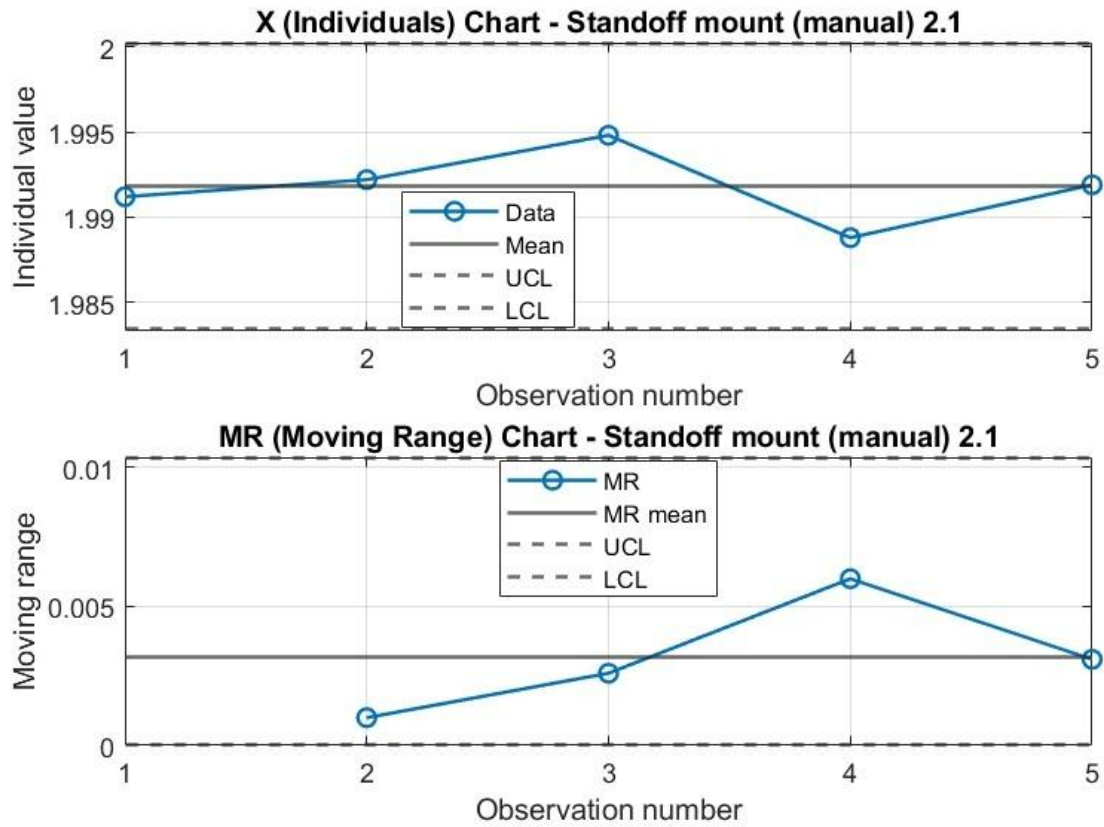


Figure 5 – Standoff Mount 2.1 Control Chart - XMR

Standoff mount (manual) - Inspection 2.2

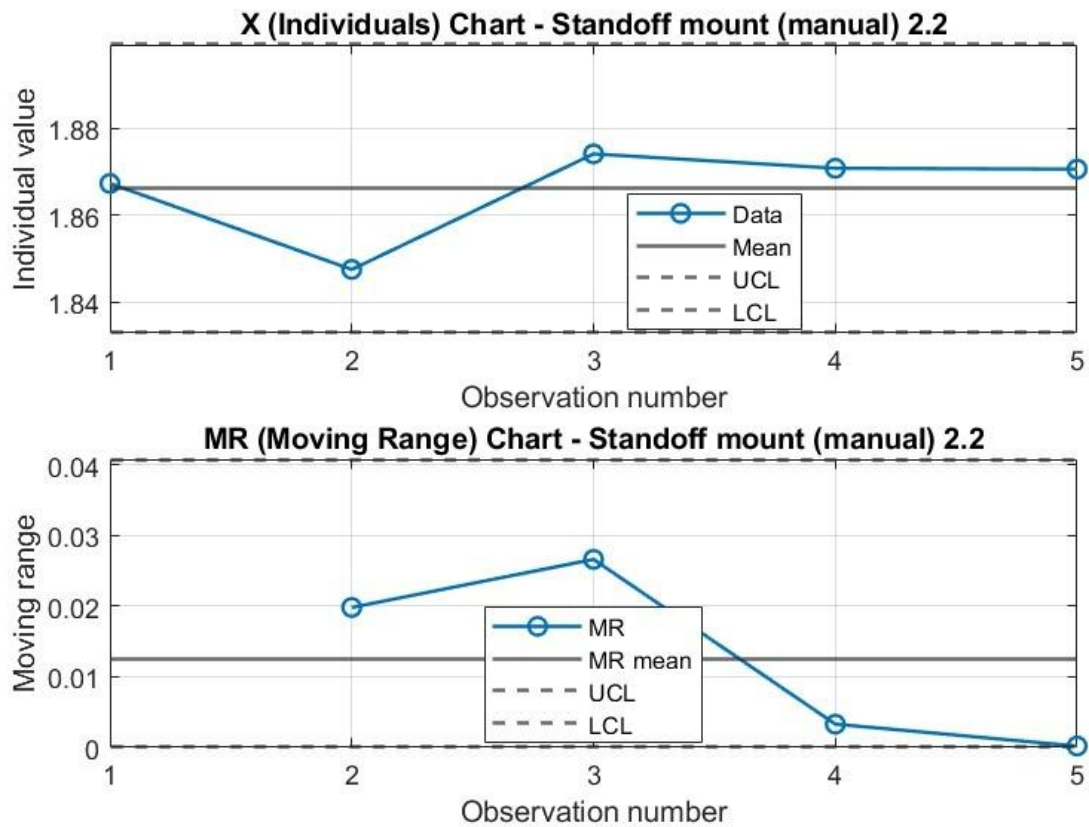


Figure 6 – Standoff Mount 2.2 Control Chart - XMR

Standoff mount (manual) - Inspection 2.3

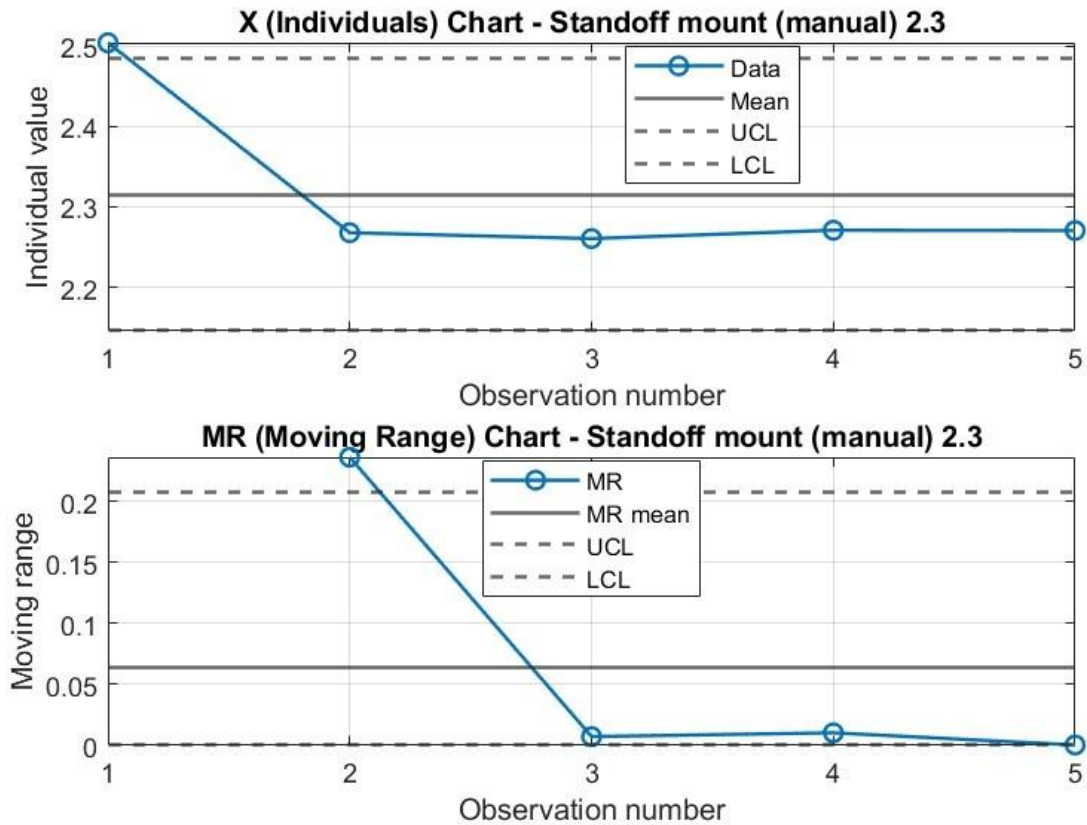


Figure 7 – Standoff Mount 2.3 Control Chart - XMR

Standoff mount (CNC) - Inspection 3.1

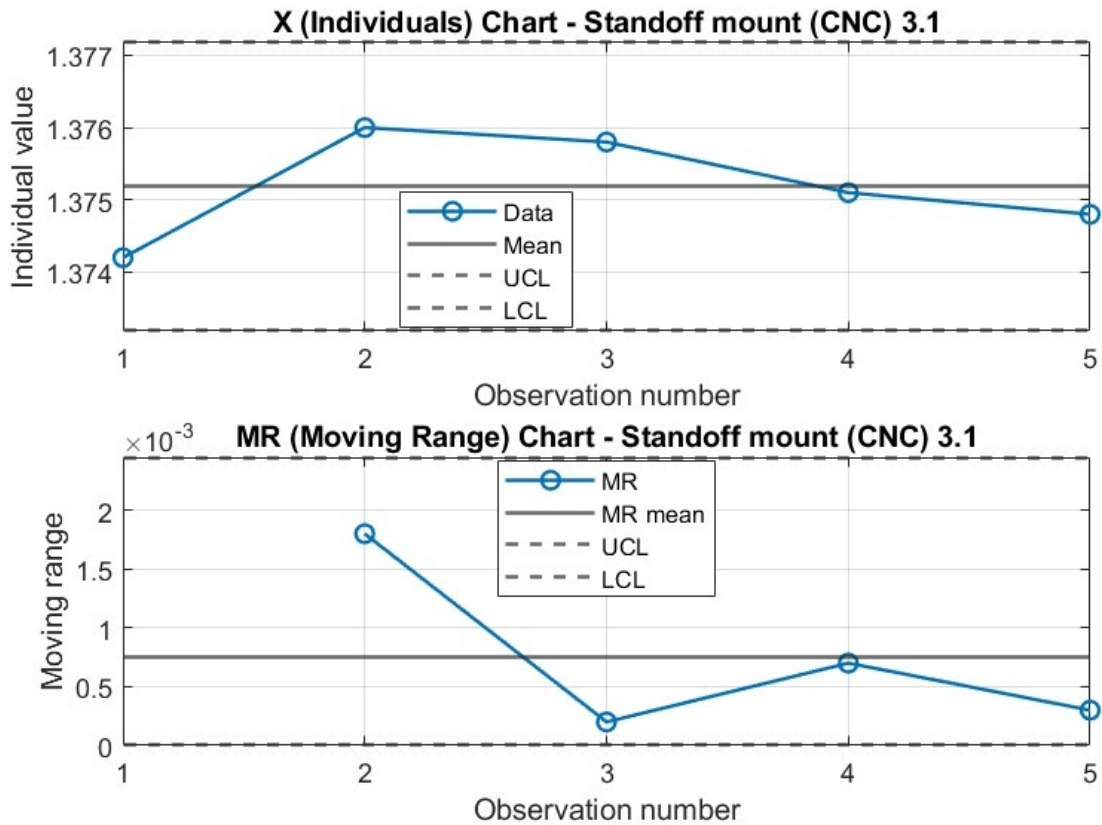


Figure 8 – Standoff Mount 3.1 Control Chart - XMR

Standoff mount (CNC) - Inspection 3.2

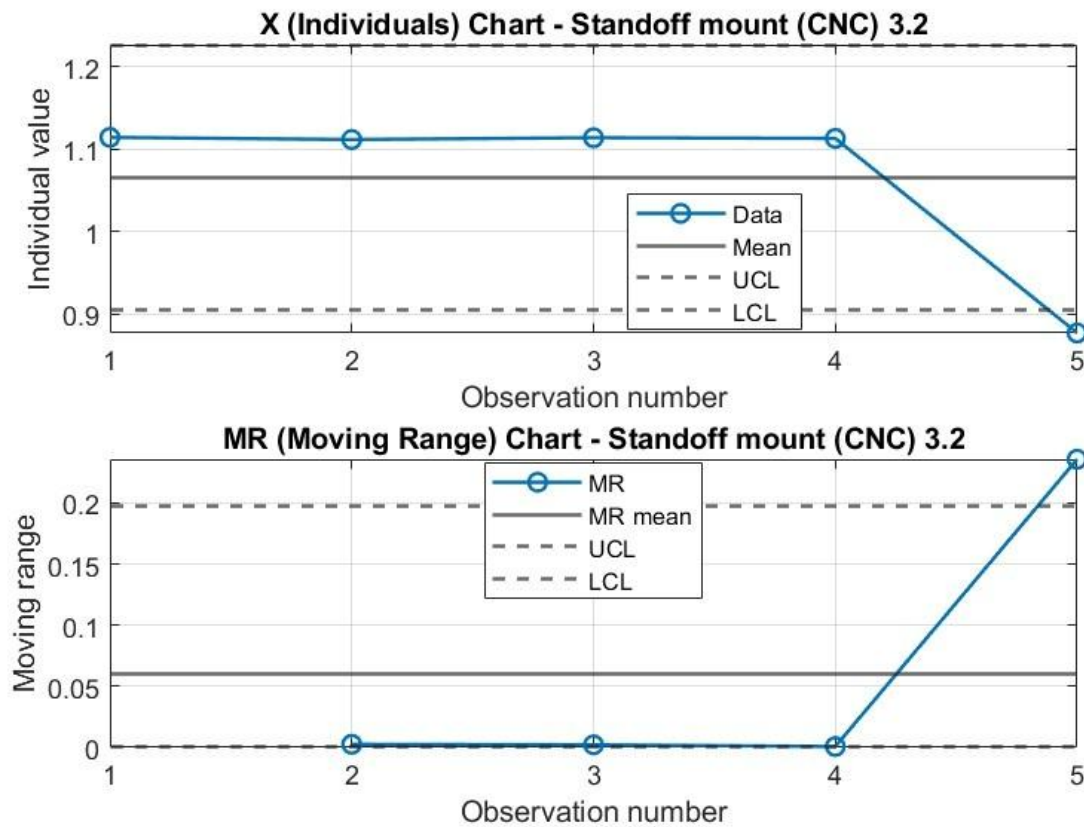


Figure 9 – Standoff Mount 3.2 Control Chart - XMR

Standoff mount (CNC) - Inspection 3.3

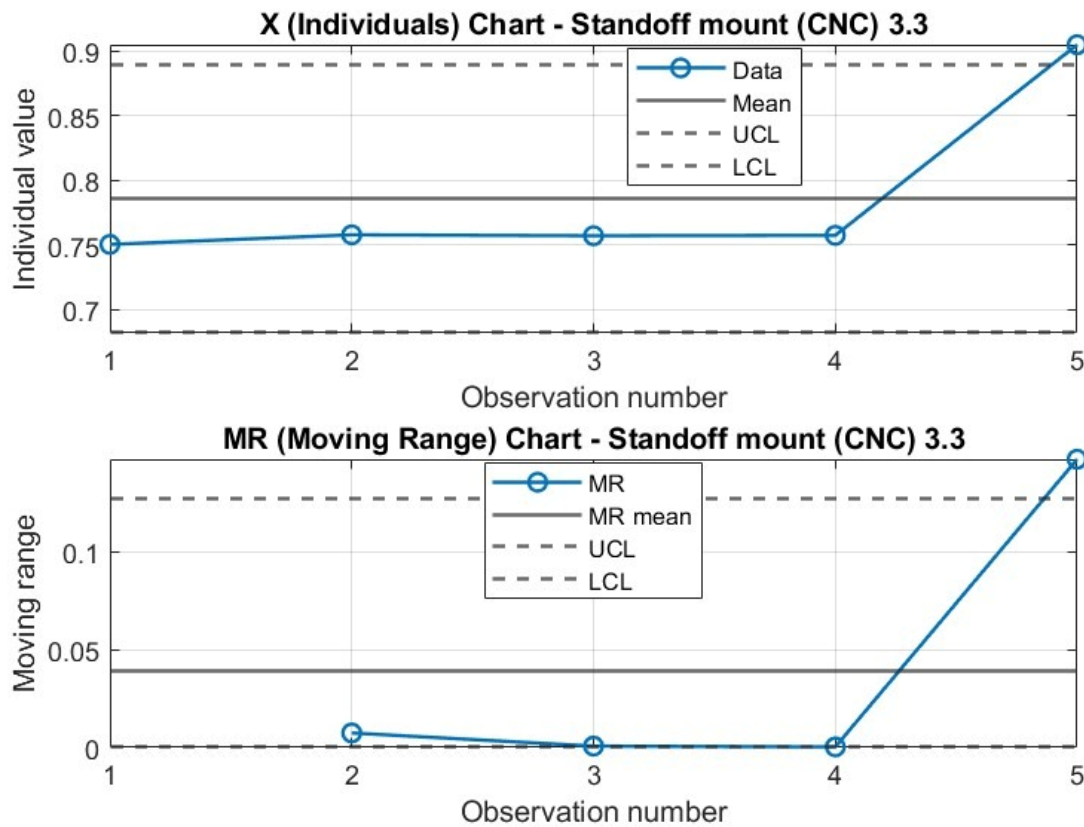


Figure 10 – Standoff Mount 3.3 Control Chart - XMR

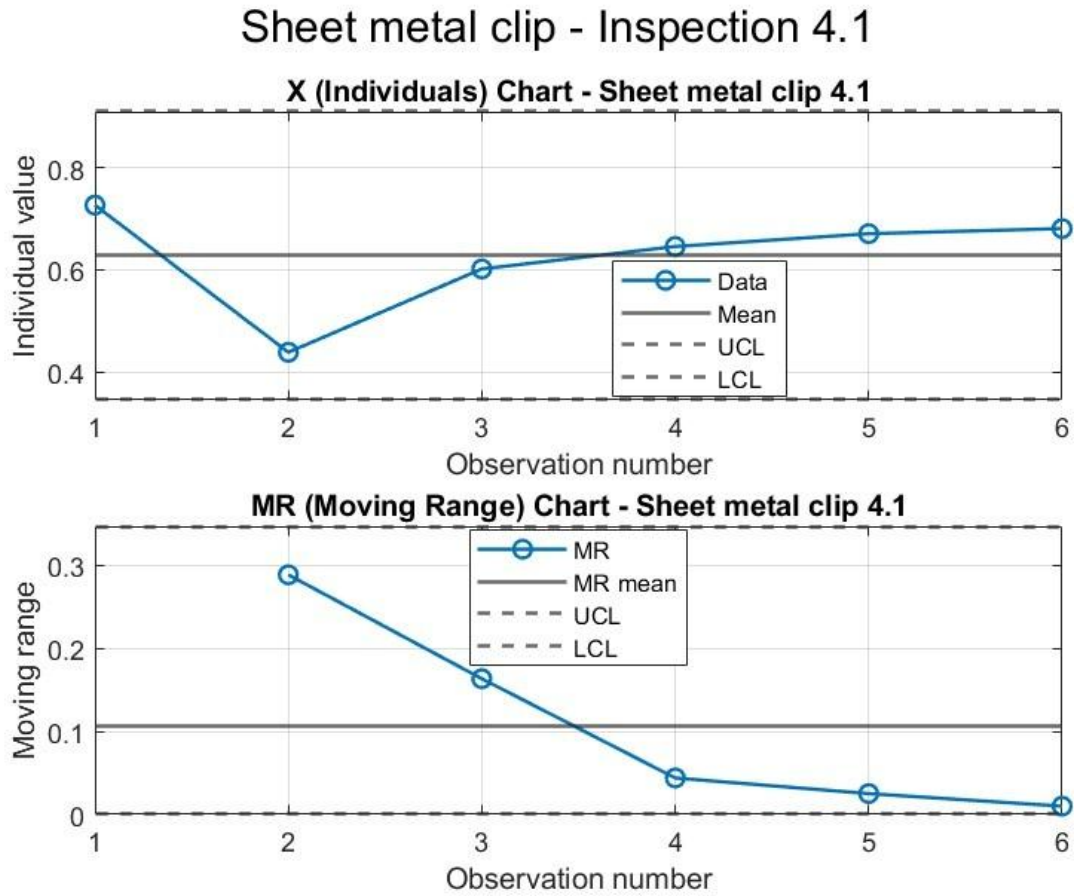


Figure 11 – Sheet Metal Clips 4.1 Control Chart - XMR

Sheet metal clip - Inspection 4.2

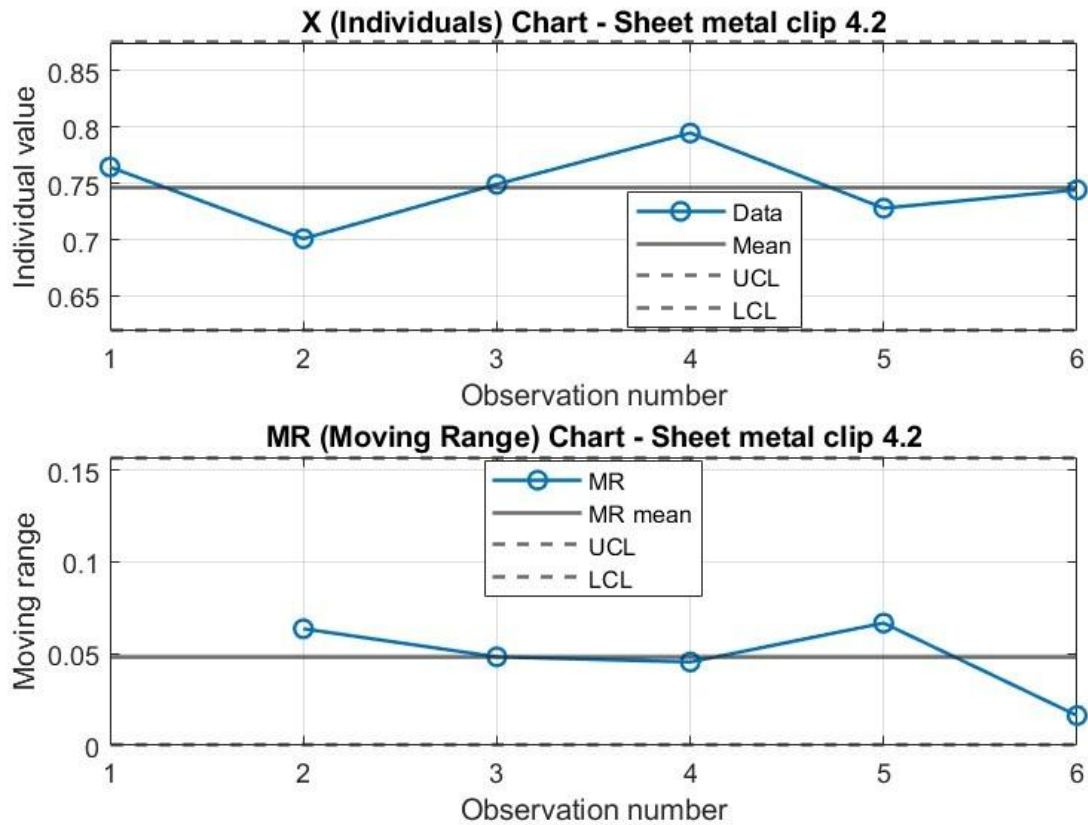


Figure 12 – Sheet Metal Clips 4.2 Control Chart - XMR

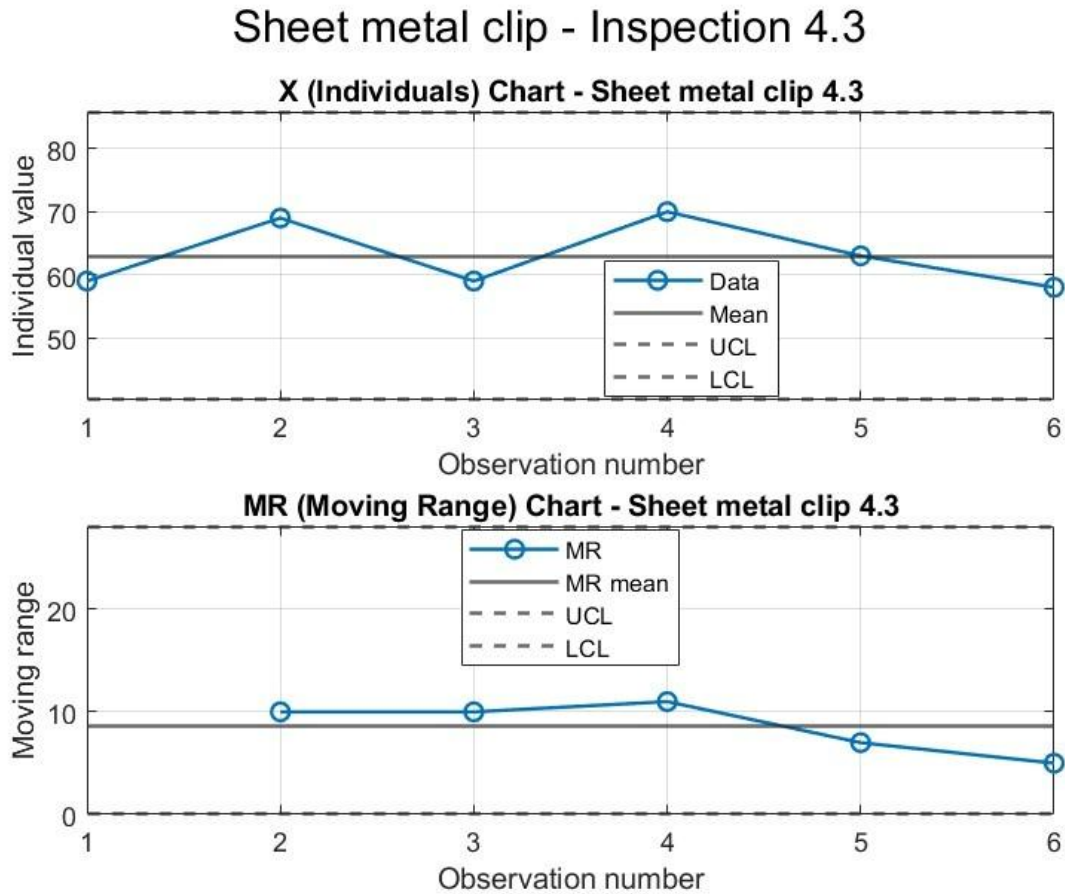


Figure 13 – Sheet Metal Clips 4.3 Control Chart - XMR

Full Inspection Reports

[CLM Full Inspection](#)

[STM Full Inspection](#)

[Clips Full Inspection](#)

[Display Plate Full Inspection](#)

[Final Assembly Full Inspection](#)

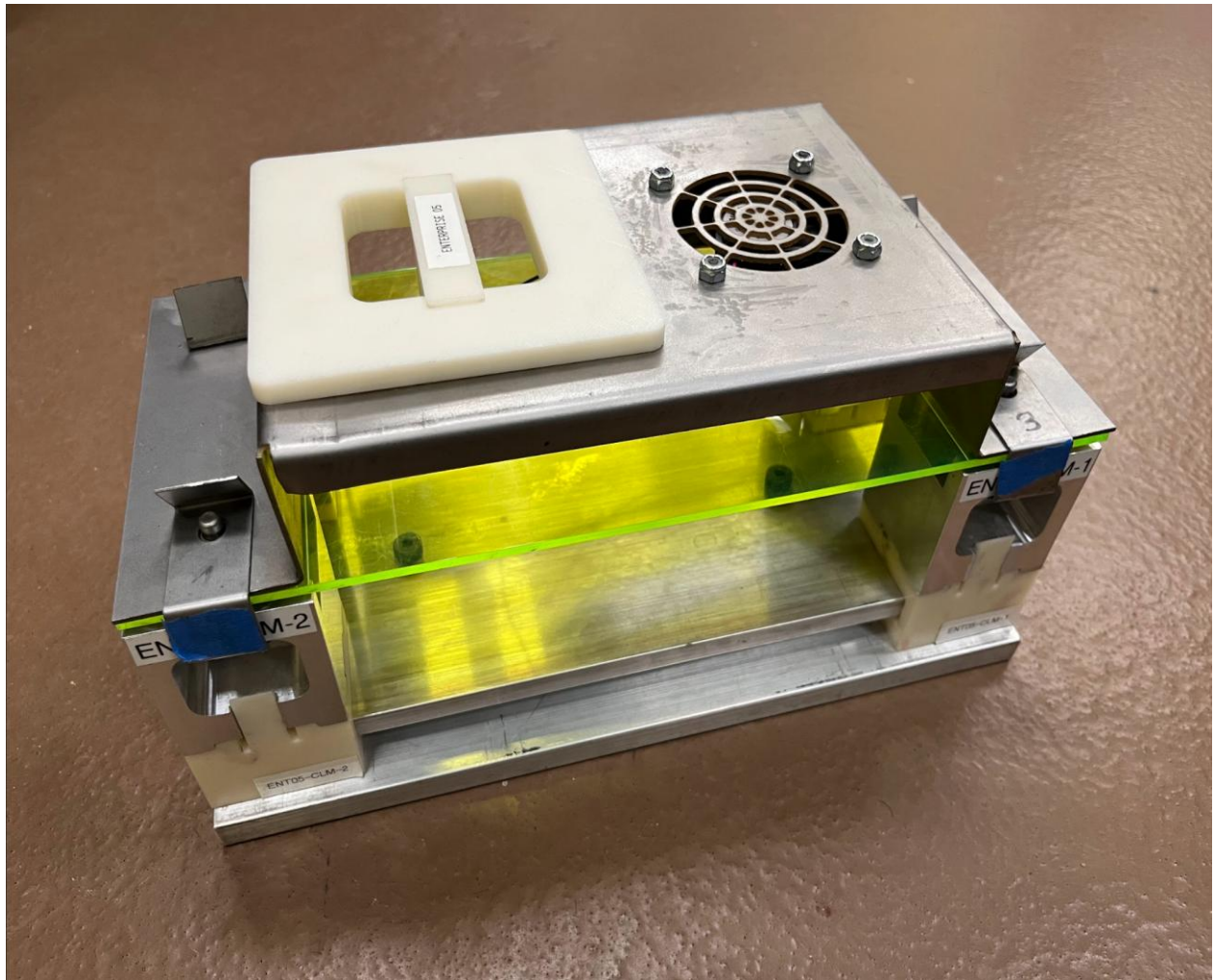


Figure 14 – Full Final Assembly

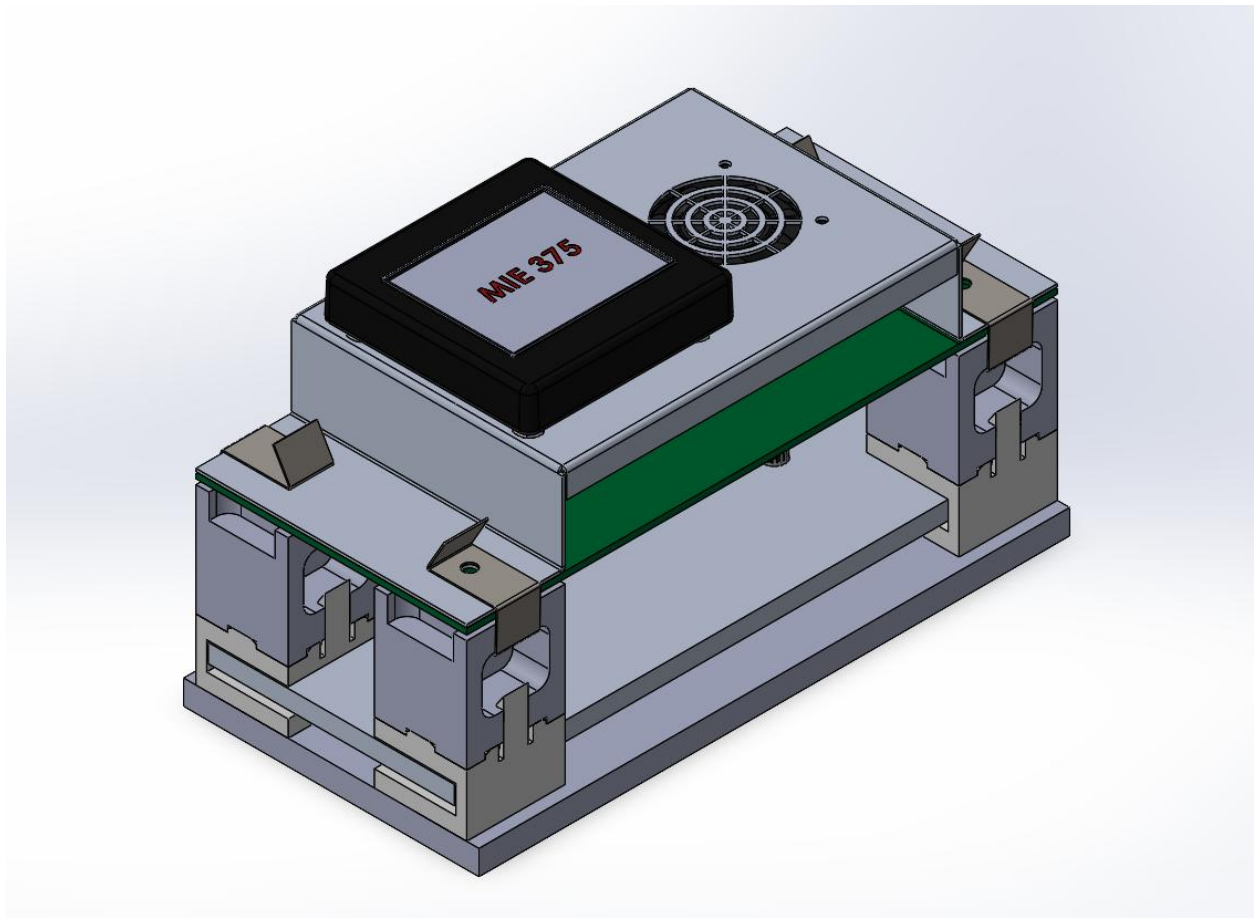


Figure 15 – Full Final CAD Assembly